A small-prime point-counting signature across the omitted curves

Date: 2026-10-24 UTC. **Question:** Under the conditional premise that the original program reached all eleven omitted prime-order curves, could an unusually *hard* Schoof–Elkies–Atkin (SEA) point count have caused them to be skipped? **Result:** The frozen optimistic Elkies-only proxy points mostly in the opposite direction. Omitted curves reach its trace-information threshold at a smaller prime in **24 of 25** same-field comparisons. P-256 block 876 is the exception, so there is no universal cutoff separating all eleven. This is a verified arithmetic pattern in the named candidate set, not a measured runtime, a historical selection rule, or an ECDLP weakness.

Why this was worth checking

Retrieved: Contemporary work on Schoof point counting distinguished Atkin and Elkies cases; the small-prime behavior can affect an implementation's workload (Morain, 1995). A later primary mathematical analysis explains why the supply of small Elkies primes matters to SEA's expected complexity, while Atkin primes also carry information (Shparlinski and Sutherland, 2014). The November 1997 X9.62 draft, §F.3.2 note 5, says orders can be computed by Schoof's algorithm and describes improvements, but does not identify Bob Reiter's point-counting implementation or its per-candidate time limits. The **user's premise that all eleven were reached remains conditional**.

Internal prior context: An earlier 10,000-degree rational-isogeny screen already computed the same Frobenius-character family and found 20 of 25 same-field pairs with *more total Elkies-or-ramified degrees* on the omitted side. That prior work also rejected a simple "few Elkies degrees caused timeout" interpretation. This round asks a narrower workload question: how far an optimistic **Elkies-only** counter must scan before enough CRT information accumulates. Its 24/25 direction is a refinement of the earlier trend, not an independent discovery or second significance test.

Frozen exact screen

The protocol, written before any curve signature was calculated, fixed the eleven omitted seed classes and eight published controls in matching fields, three metrics, controls, and a 60-second wall stop. The pre-run amendment corrects a prime-count typo: there are 167 odd primes through 997, not 166. For each archived `p,n`, we independently reconstructed $t=p+1-n$ and $\Delta=t^2-4p$. For every odd prime $\ell \le 997$, Δ a quadratic residue modulo ℓ is called Elkies, a nonresidue Atkin, and zero ramified. We multiplied the Elkies primes in order and recorded the first ℓ where $M^2>16p$, an optimistic CRT threshold sufficient to distinguish a trace in Hasse's interval if each contributing residue were known. It is **not** an SEA operation count: modular-polynomial cost rises with ℓ, actual software may use Atkin information, and neither original code nor hardware is known.

Internally verified: The first run's raw record and fully logged second run's raw record are byte-identical (SHA-256 `4a0df1de260e4533253ef591f7b084b73e86fac5e03558ec3609081c750f48fe`). The independent Sage verification recomputed all **3,173** Legendre symbols and all 19 thresholds from the archived `p,n` values, checked Hasse bounds and exact UIDs, and passed under SageMath 10.10.rc0. The synthetic residue and strict-boundary controls passed. The run receipt records

Python, platform, source head, dirty state and wall/CPU time; stdout and stderr are retained. RUN_NOTE.md corrects two misleading directory-time suffixes and identifies the first run's absent console logs. The study's filename correction changes no arithmetic.

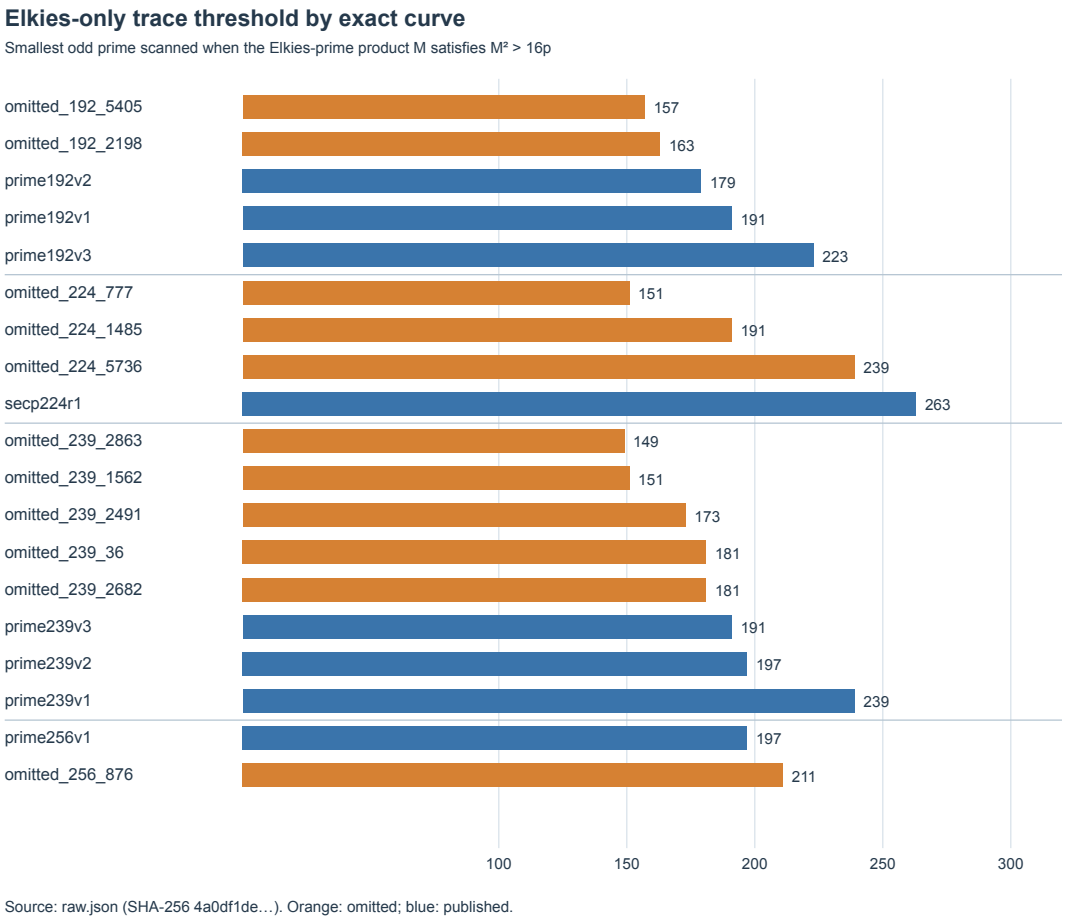


Figure 1: Exact small-prime cutoffs for all 19 curves

Orange bars are omitted seed classes; blue bars are published controls. Smaller is a lower Elkies-only trace threshold. Source: the exact raw JSON; editable chart source: render.py.

Field	Omitted cutoffs ℓ	Published cutoffs ℓ	Omitted lower in within-field pairs
P-192	157, 163	179, 191, 223	6 / 6
P-224	151, 191, 239	263	3 / 3
ANSI 239-bit	149, 151, 173, 181, 181	191, 197, 239	15 / 15
P-256	211	197	0 / 1

Derived: Across the 25 pair comparisons, the omitted cutoff is lower in 24 and higher in one, with no ties. The two prelisted small-prime density metrics point the same general way but less cleanly: omitted curves have more Elkies primes through 101 in 21 of 25 pairs, and through 251 in 20 pairs, with two ties. Within P-192, P-224 and ANSI 239-bit, the cutoff values perfectly order the named published and omitted candidates; the P-256 pair reverses that ordering. The maximum published cutoff is 263, while the minimum omitted cutoff is 149, so no global rule of the form "reject if the Elkies-only cutoff exceeds `c`" separates the labels. More pointedly, the omission side is usually *easier* under this proxy, opposing an "old point counter ran out of time on hard omissions" story.

The post-result reallocation counts all **4,480** possible within-field label allocations preserving published counts. Four yield at least the observed 24/25 pair score. This is **descriptive only**: the historical labels are not randomized, candidate streams are selected and dependent, and this particular proxy was chosen after many prior features were examined. The ratio is not a probability of hidden selection or a general statistical significance claim.

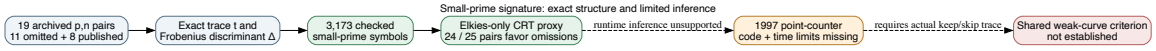


Figure 2: Inference boundary

Solid arrows are exact input and arithmetic checks; dashed arrows require unavailable program traces or a mechanism. Editable source: inference.dot.

What remains open

A true small-Elkies-based *security* concern would need to show how those small rational isogenies or their resulting structure lower unknown-scalar ECDLP cost on an exact omitted prime-order subgroup, with construction and evaluation costs and matched published controls. The verified pattern does not do that. The P-256 reversal prevents this exact scalar proxy from being a common all-eleven weakness even as an arithmetic label rule. A broader, predeclared criterion could still exist; one can fit many retrospective conditions to nineteen labeled curves.

The best process test remains the original point-counter and its per-candidate logs, followed by a source-identical bounded replay. This screen sharpens the previously reported small-Elkies trend for the May 1997 and April 1998 groups; it does not establish why a published curve was chosen. The selection problem remains open. I checked the existing seed-selection and canonical ECDLP graph sources; this round creates no verified attack, isogeny map or cost edge, so none requires a factual update. Provenance: original documents `retrieved`; p,n and prior certifications `internal`; signatures and comparisons `independently checked/derived`; process and weakness interpretations `proposed`.

Reproduction

The exact commands, input digests and environment are in the logged run receipt and protocol. Run run.py and the independent Sage verify.py into fresh output paths. explore_labels.py is a *post-result* diagnostic outside the frozen primary metrics. Existing runs are not overwritten.